PR 1749 review correction

2026-10-19 · implementation and evidence correction

Target stream correction · PR #1749 · 2026-10-19

Historical: draw x ⟶ draw sign ⟶ attempt lift

Corrected: draw x ⟶ attempt lift ⟶ draw sign only if lift succeeds

Frozen records remain unchanged; corrected-protocol evidence is absent.

Native ecbench admission and paired IDC1 certificate remain open.

Historical evidence boundary

The runner at a4daad2d441d157a32c36bd144fdf4478e7fa690 drew a sign before checking whether the abscissa lifted. PROTOCOL.md requires the sign after a successful lift. The source is corrected, with a regression fixing the third F13 target to (5,7). No replacement experiment occurred. Frozen raw-v4.tsv, summaries, figures, PDFs and receipts are preserved; they do not establish compliance with the corrected target stream.

Catalogue repair

A normal main merge combines identities and regenerates derived catalogue, cover, trait, transfer, leaderboard, browser and dashboard files. The complete N73 EC1 record in research/ic_single_target_20260930/curve_ids.json was omitted by a filename-restricted harvester. The generator now reads complete records from curve_ids.json and curve_records.json and rejects a declared UID that disagrees with its field and curve. Frozen inputs are unchanged. Regression tests cover retention and corrupted-UID rejection.

Unresolved merge gates

The denominator-cleared map equation now has a native IDC1 emitter and checked Lean 4.34.1 ring proofs: degreeThreeCleared and degreeThreeCurveEquationCleared in CubicAutomorphism.lean. Both sides expand to $x^9 - 15bx^6 + 48b^2x^3 + 64b^3$. Their axioms are propext, Classical.choice and Quot.sound. The IDC1 field is $2^{61} - 1$; the Lean statements apply to any commutative ring in the stated interface. Neither certifies the separate finite-group decomposition. Certificate: IDC1h95b8d6aaf113c1db; 32 certificate points and 32 independent checker-stream points. The proof-source hash and axiom receipt are linked in REVIEW-20261019.md.

The paired comparison needs a new versioned native ecbench protocol, sealed session, verified answers, controls and audit. Its paired isogeny/scalar decomposition also requires the IDC1 identity certificate and faithful proof bridge. Other torsion-study certificates do not certify this identity. Applicable exact-head CI must pass. Recording these gaps does not make the PR ready to merge.

Sources and graph impact

REVIEW-20261019.md links the sources, commands and existing obstruction record, problems/nonrational-isogeny-fiber-dlp-cost-gap.md. The corrected source is examples/isogeny_fiber_pair_20261012.rs. Canonical indexes are regenerated without changing measured values, incumbents or reference costs. Historical paired figures and benchmark mirrors are preserved and cannot establish a newly admitted comparison. This diagram records control flow; this PDF reports no new measurements.